

PAWAN RAGHUWANSHI

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Education

Shri G.S. Institute of Technology and Science (SGSITS), Indore

2023 – 2027 (Expected)

Bachelor of Engineering in Electrical Engineering

Indore, India

Technical Skills

Languages: C++ (Primary), Python, Java, JavaScript, C, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, DBMS, Complexity Analysis, System Design, Distributed Systems

Backend & Systems: Node.js, Express.js, RESTful APIs, Service-Oriented Architecture, Socket.IO (event-driven/real-time systems), JWT Authentication, RBAC

Databases: MongoDB, Mongoose ODM, Query Optimization, Indexing

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Tools: Git, GitHub, Vite, Agile/Scrum workflow, API Integration

Experience

Technical Lead & Full-Stack Developer

April 2026 – July 2026

SGSITS, Dept. of Pharmacy (Media Cell) – Internship, Full-Stack Development

Indore, India

- Led a cross-functional team of 4 engineers through the full software lifecycle – system design, implementation, and deployment – to build **ChemInventory**, a distributed inventory and request-management platform serving 2,000+ students and staff, owning end-to-end technical decision-making.
- Designed a multi-role, fault-tolerant access-control system with audit-trail logging to support secure, traceable request workflows across a catalogue of 300+ items, prioritizing data consistency under concurrent multi-user access.
- Built a real-time monitoring module with automated low-stock alerts and reorder triggers, applying event-driven design to keep the system responsive and scalable as usage grew.
- Developed modular, service-oriented RESTful APIs with centralized validation and structured error handling, working in an agile environment with iterative delivery and code review cycles across the team.

Projects

RasayanFlow – Laboratory Inventory & Request Management System

[Link](#) | [Github](#)

React.js, Node.js, Express.js, MongoDB, Socket.IO, JWT

- Engineered a full-stack MERN application for inventory tracking and real-time request management, designing the data model and API layer to support concurrent multi-user access via Socket.IO without polling.
- Implemented a Role-Based Access Control (RBAC) system with JWT authentication and Express.js middleware to secure multi-role workflows (admin, staff, student), reasoning explicitly about auth and authorization edge cases.
- Integrated PubChem and PubMed APIs to enrich dataset metadata and improve search relevance, and applied MongoDB indexing and query optimization alongside React lazy loading/code splitting to improve system performance at scale.
- Built modular, testable RESTful APIs following service-oriented design, with centralized validation and structured error handling across all endpoints.

Full-Stack Blogging Platform

[Github](#)

Node.js, Express.js, MongoDB, EJS, Bootstrap

- Built a full-stack content management platform supporting 50+ users and 100+ published posts, with session-based authentication, server-side rendering, and scalable MongoDB data persistence using Mongoose schema design.
- Designed RESTful APIs with query-level indexing across the post and user collections, input validation, and centralized error handling following modular, service-oriented backend architecture.

Achievements & Certifications

- Ranked in the **Top 1%** of participants in **Flipkart Grid 8.0**, a national-level engineering and problem-solving competition.
- Solved **300+ Data Structures & Algorithms** problems on LeetCode and GeeksforGeeks, building strong fundamentals in algorithmic problem solving and complexity analysis for scalable system design.
- **McKinsey Forward Program Fellow** – Completed structured problem-solving and business-thinking training, developing the ability to reason abstractly about ambiguous, broadly-defined problems.
- **NPTEL Database Management Systems (IIT Kharagpur)** – Elite + Top 5% nationally; in-depth knowledge of relational databases, query optimization, and scalable data management.
- **The Complete Web Development Bootcamp** (Udemy, Dr. Angela Yu) – Project-based full-stack curriculum spanning HTML/CSS/JS, React, Node.js, Express, and MongoDB.